Is Nutrient Limitation Synchronous Across Forest Trees, Understory, and Soil Microbes? A Global Meta-Analysis of N and P Experiments

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Background and Ecological Problem

Knowledge at hand

- Global change biology focuses heavily on canopy tree or microbial responses to nitrogen (N) and phosphorus (P) deposition (Cui et al., 2025; Du et al., 2020, 2024; Li et al., 2021)..

Knowledge Gap

1. Ecosystems are multi-trophic. Canopy trees, understory plants, and soil microbial biomass (C:N:P) compete for the same nutrient pools.
2. Yet most studies evaluate nutrient limitation in one compartment in isolation.
3. The core question is do canopy trees, understory plants, and soil microbes become N, P or N+P limited synchronously at the same threshold, or are there stoichiometric lags and competitive trade-offs between them?

In general, I would like to conduct the meta-analysis as done in Sánchez-Tójar and D'Amelio (2026)

PICO Framework and Objectives according to Sánchez-Tójar and D'Amelio (2026)

- Population (P): Natural forest ecosystems globally (boreal, temperate, tropical).
- Intervention (I): Field nutrient additions (N, P or N+P).
- Comparator (C): Control plots.
- Outcomes (O): Concurrent effect sizes (ln RR (response ratio)) for primary growth or biomass across three distinct compartments

    1. Canopy Trees (stem diameter or growth or leaf biomass).
    2. Understory Vegetation (community cover or biomass).
    3. Microbial Biomass (C, N, P biomass or microbial respiration).

Key Methodological Challenges Alongside Others

- Challenge 1: Non-Independence and Hierarchy
  - Multiple compartments (tree, understory, microbe) and treatments (N, P or N+P) are not measured within the exact same experimental setup or sites most times.
  - Planned Approach: Fitting Multi-Level Meta-Analysis models in metafor (rma.mv) nesting Observation → Compartment → Study → Site

- Challenge 2: Publication Bias and Small-Sample Effects
  - Possible selective reporting of positive growth responses.
  - Planned approach is to make a funnel plot symmetry, Egger's tests, and P-curve analyses across sub-groups.

- Challenge 3: Pre-Registration
  - Finalising a rigorous Open Science Framework (OSF) protocol before full screening, time to do that and the mata-analysis within my limited timerame.

Connection to My Experiments and Expected Impact

- Local Field Site Integration
  - Directly informs my field experimental manipulations at Aspurz and Garde (Navarre, Spain), where we measure N:P critical ratios across trees, roots, understory, and soil microbes.

- Broader Significance
  - Provides global parameterisation for multi-trophic ecosystem simulation models.

- Target Publication
  - Global Change Biology or Ecology Letters.

References

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